

Student-Created Problem Set Mission

Warm-up and worked example

Name: _____

Date: _____

PAGE 1 | WEEK 9, DAY 5

YOUR MISSION Create, solve, revise, and explain original mathematical problems.

SMART MOVE Draw a simpler case and organize what you know.

Math Talk

What makes a math problem interesting rather than merely long?

My idea:

New Words

represent	deduce	assumption
My meaning or picture:	My meaning or picture:	My meaning or picture:

Worked Example

Do this example with your teacher. Say what changes and what stays the same.

1. One two-step money problem.
2. One multiplication or division problem.

What the example shows: _____

Student-Created Problem Set Mission

Learn it and show your thinking

Name: _____

Date: _____

PAGE 2 | WEEK 9, DAY 5

YOUR MISSION Follow the learning path, then explain the big idea.

SMART MOVE Draw before asking for a rule.

[] **Step 1:** One two-step money problem.

[] **Step 2:** One multiplication or division problem.

[] **Step 3:** One data question.

[] **Step 4:** One challenge problem with more than one possible answer.

My model, drawing, or organized work:

The big idea in my own words:

Student-Created Problem Set Mission

Practice lab

Name: _____

Date: _____

PAGE 3 | WEEK 9, DAY 5

YOUR MISSION Use a model, equation, or explanation for every task.

SMART MOVE Check whether your answer is reasonable.

1. A number is even, greater than 30, less than 50, and its digits total 10. Find all possibilities.

2. How many different two-digit numbers can be made with 2, 5, and 8 without repeating a digit?

3. Estimate how many piano keys are in 12 classrooms if each has about 1 piano. State an assumption.

4. Find a pattern rule for 3, 7, 11, 15, ... and give the tenth term.

5. A farmer has chickens and goats with 12 heads and 34 legs. How many of each?

6. Choose one answer above and prove that no case was missed.

Choose one answer and prove it another way:

Student-Created Problem Set Mission

Challenge and final check

Name: _____

Date: _____

PAGE 4 | WEEK 9, DAY 5

YOUR MISSION Try an unfamiliar problem and defend your reasoning.

SMART MOVE A wrong first idea can still lead to a strong solution.

Challenge Mission

Create a problem that appears difficult but has an elegant shortcut.

My plan:

My solution and proof:

Final Check

Write the intended strategy beneath each problem.

☐ I can teach it

☐ I need practice

☐ I need help

Student-Created Problem Set Mission

Reusable math tool

Name: _____

Date: _____

PAGE 5 | WEEK 9, DAY 5

YOUR MISSION Use this page to draw, model, organize, and check.

SMART MOVE Draw a simpler case and organize what you know.

Problem-Solving Organizer

What I know

What I need to find

My model or table

My equations

My check and explanation
